Fick's Laws and Transport Equations

$$J(x,t) = -D \nabla C(x,t) \tag{1}$$

$$J_x(x,t) = -D \partial C/\partial x \tag{2}$$

$$J_adv = C\ u \tag{3}$$

$$J_total = C\ u - D \nabla C \tag{4}$$

$$\dot{n} = \int_s J \cdot n\ dA \tag{5}$$

$$d/dt \int_v C\ dV = -\int \partial_v J \cdot n\ dA + \int_v s\ dV \tag{6}$$

$$\partial C/\partial t + \nabla \cdot J = s \tag{7}$$

$$\partial C/\partial t = \nabla \cdot (D \nabla C) + s \tag{8}$$

$$\partial C/\partial t = D \nabla^2 C \tag{9}$$

$$\partial C/\partial t + u \cdot \nabla C = D \nabla^2 C + s \tag{10}$$

$$J_x = -D\ [C(b) - C(a)] / L \tag{11}$$

$$n \cdot \nabla C = 0 \tag{12}$$

$$s(x,t) = \dot{n}_inj(t)\ \delta(x - x_0) \tag{13}$$

$$C(x,t) = M / (4\pi Dt)\text{^}(3/2)\ \exp[-|x - x_0|\text{^}2/(4Dt)] \tag{14}$$

Variable Key

C	solute concentration
J	diffusive flux vector
Jx	x-direction diffusive flux
J_adv	advective solute flux
J_total	total solute flux
D	molecular diffusivity
u	fluid velocity vector
n	outward unit normal vector
n-dot	molar flow rate
s	volumetric solute source or sink rate
V	control volume
partial V	boundary surface of V
S	surface used to evaluate flux
A	surface area
L	one-dimensional diffusion length
x, x0	position and injection position
M	instantaneously injected solute mass
t	time
delta	Dirac delta function
nabla	spatial-gradient operator